

163. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.1 Overview

This clause specifies the 100GBASE-KR1 PMD, the 200GBASE-KR2 PMD, the 400GBASE-KR4 PMD, and the baseband medium. The specifications for the three PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are two associated annexes. Annex 163A provides measurement methods and test points for backplane and chip-to-chip interfaces. Annex 163B provides information on an example test fixture.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 163–1, to the appropriate sublayers (as specified in Table 163–1, Table 163–2, and Table 163–3), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 163–1—Physical Layer clauses associated with the 100GBASE-KR1 PMD

Associated clause	100GBASE-KR1
81—RS	Required
81—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC	Required
152—Inverse RS-FEC	Optional ^b
161—RS-FEC-Int	Required
83—PMA for 100GBASE-R	Optional ^c
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135F—100GAUI-2 C2C	Optional
120F—100GAUI-1 C2C	Optional
73—Auto-Negotiation	Required

^a The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^b Inverse RS-FEC is required to convert between RS-FEC and RS-FEC-Int (see 152.1.2).

^c PMA for 100GBASE-R is required when either CAUI-10 or CAUI-4 is used.

Table 163–2—Physical Layer clauses associated with the 200GBASE-KR2 PMD

Associated clause	200GBASE-KR2
117—RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—PCS for 200GBASE-R	Required
120—PMA for 200GBASE-R	Required
120B—200GAUI-8 C2C	Optional
120D—200GAUI-4 C2C	Optional
120F—200GAUI-2 C2C	Optional
73—Auto-Negotiation	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

Table 163–3—Physical Layer clauses associated with the 400GBASE-KR4 PMD

Associated clause	400GBASE-KR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—PCS for 400GBASE-R	Required
120—PMA for 400GBASE-R	Required
120B—400GAUI-16 C2C	Optional
120D—400GAUI-8 C2C	Optional
120F—400GAUI-4 C2C	Optional
73—Auto-Negotiation	Required

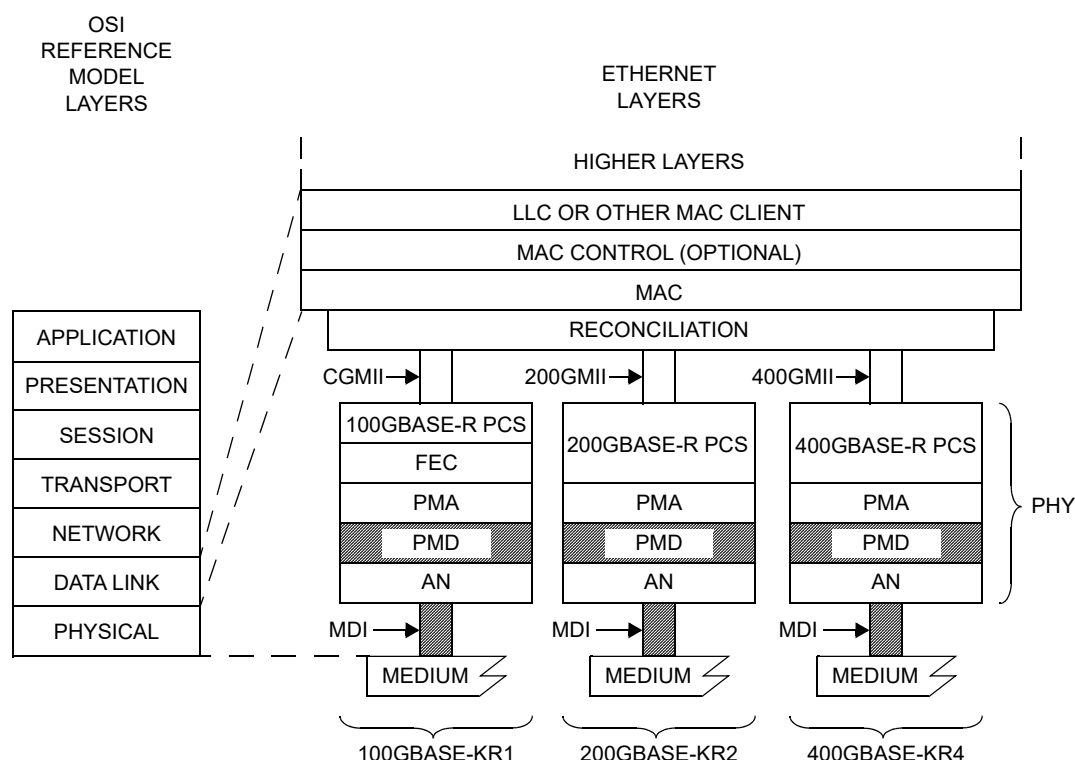
^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

For the 100GBASE-KR1 PHY, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-10} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 10^{-10} after processing by the FEC (Clause 91 or Clause 161) and the PCS (Clause 82) if there are negligible errors due to other electrical interfaces (100GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 10^{-10} .

For the 200GBASE-KR2 and 400GBASE-KR4 PHYs, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 9.2×10^{-13} after processing by the PCS (Clause 119) if there are negligible errors due to other electrical interfaces (200GAUI-n or 400GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 9.2×10^{-13} .

A compliant input signal is a transmitter output of a compliant PHY that has passed through a compliant channel.

Figure 163–1 shows the relationship of the PMD and MDI to the ISO/IEC Open System Interconnection (OSI) reference model.



200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
AN = AUTO-NEGOTIATION
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL

MDI = MEDIUM DEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 163–1—100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

163.2 Conventions

Clause 163 describes three PMDs, 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, which have one, two, and four lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-KR1, $n = 2$ for 200GBASE-KR2, and $n = 4$ for 400GBASE-KR4.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-KR1 PMD, 200GBASE-KR2 PMD, or 400GBASE-KR4 PMD.

Within this clause, the term FEC refers to either Clause 91 RS-FEC or the Clause 161 RS-FEC-Int for 100GBASE-KR1, or the RS-FEC within the Clause 119 PCS for 200GBASE-KR2 and 400GBASE-KR4.

163.3 PMD service interfaces

The service interfaces of the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PMDs are identical to those of the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs, respectively (see 162.3).

163.4 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive AN_LINK.indication defined in 73.9. The requirements are the same as those specified in 162.4.

163.5 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN including the medium in one direction shall be no more than the maximum delays listed in Table 163–4. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 163–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-KR1	4 096	8	40.96
200GBASE-KR2	8 192	16	40.96
400GBASE-KR4	16 384	32	40.96

^a One bit time is equal to 10 ps for 100GBASE-KR1, 5 ps for 200GBASE-KR2, and 2.5 ps for 400GBASE-KR4. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-KR1, 2.56 ns for 200GBASE-KR2, and 1.28 ns for 400GBASE-KR4. (See 31B.2 for the definition of pause_quanta.)

Descriptions of overall system delay constraints can be found in 80.4 for 100GBASE-KR1 and in 116.4 for 200GBASE-KR2 and 400GBASE-KR4.

163.6 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

163.6.1 Skew Constraints for 100GBASE-KR1

Skew and Skew Variation are defined in 80.5 and specified at the points SP1 to SP7 shown in [Figure 80–9](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by [135.5.3.5](#). Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 80.5. The measurements of Skew and Skew Variation are defined in [89.7.2](#).

163.6.2 Skew Constraints for 200GBASE-KR2 and 400GBASE-KR4

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in [Figure 116–5](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

163.7 PMD MDIO function mapping

The PMD MDIO function mapping requirements for the PMDs defined in this clause are identical to those of 162.7.

163.8 PMD functional specifications

163.8.1 Link block diagram

One direction of a 100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link is shown in Figure 163–2. The PMDs on both ends of the link have connected ground references.

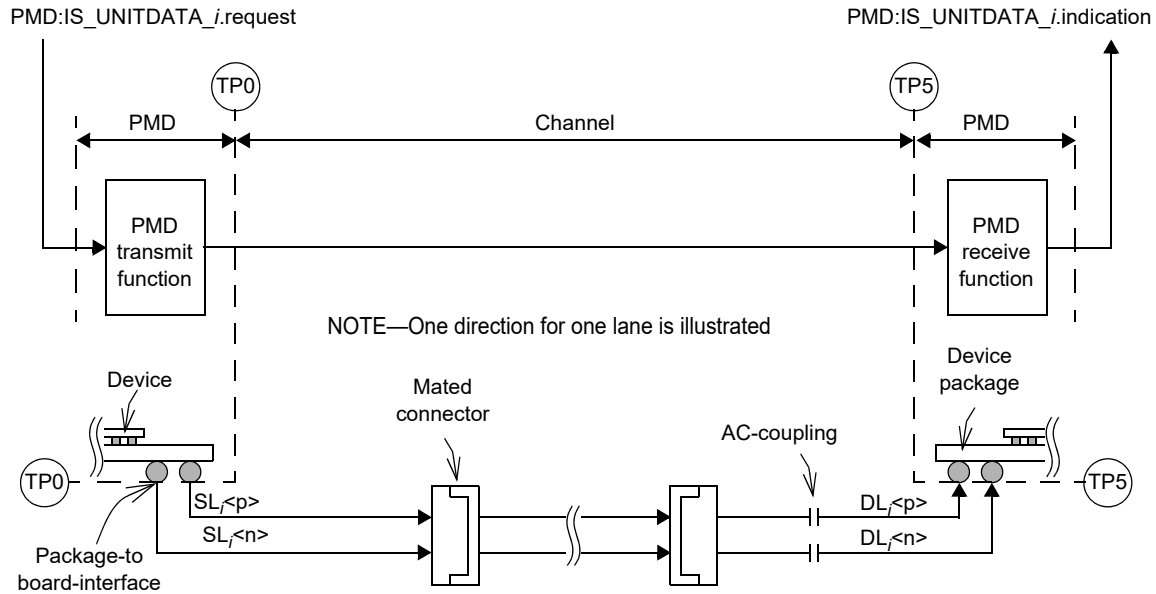


Figure 163–2—100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link

163.8.2 PMD transmit function

The PMD transmit function specification is identical to that of 162.8.2 with the exception that electrical signals are delivered to the MDI, according to the transmit electrical specifications in 163.9.2.

163.8.3 PMD receive function

The PMD receive function specification is identical to that of 162.8.3 with the exception that electrical signals are received from the MDI, according to the receive electrical specifications in 163.9.3.

163.8.4 PMD global signal detect function

The PMD global signal detect function specification is identical to that of 162.8.4.

163.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function specification is identical to that of 162.8.5.

163.8.6 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional. Its specification is identical to that of 162.8.6.

163.8.7 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional. Its specification is identical to that of 162.8.7.

163.8.8 PMD fault function

The PMD fault function specification is identical to that of 162.8.8

163.8.9 PMD transmit fault function (optional)

The PMD transmit fault function is optional. Its specification is identical to that of 162.8.9.

163.8.10 PMD receive fault function (optional)

The PMD receive fault function is optional. Its specification is identical to that of 162.8.10.

163.8.11 PMD control function

The PMD control function specification is identical to that of 162.8.11.

163.9 Electrical characteristics

163.9.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

163.9.2 Transmitter characteristics

The transmitter shall meet the specifications given in Table 163–5. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

Table 163–5—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		$53.125 \pm 50 \text{ ppm}^a$	GBd
Differential pk-pk voltage (max) ^b Transmitter disabled Transmitter enabled	93.8.1.3	30 1200	mV mV
DC common-mode voltage (max) ^b	93.8.1.3	1	V
DC common-mode voltage (min) ^b	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	162.9.4.4	30	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	163.9.2.6	15	dB
Difference effective return loss, $dERL$ (min)	163.9.2.2	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	163.9.2.3	3.25	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163.9.2.5	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 163–5—Summary of transmitter specifications at TP0v (*continued*)

Parameter	Reference	Value	Units
Transmitter waveform			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at minimum state for $c(-3)$ (max)	162.9.4.1.5	−0.06	—
value at maximum state for $c(-2)$ (min)	162.9.4.1.5	0.12	—
value at minimum state for $c(-1)$ (max)	162.9.4.1.5	−0.34	—
value at minimum state for $c(0)$ (max)	162.9.4.1.5	0.5	—
value at minimum state for $c(1)$ (max)	162.9.4.1.5	−0.2	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	162.9.4.3	28	dB
Jitter (max) ^c			
J_{RMS}	162.9.4.7	0.023	UI
J_{3u03}	162.9.4.7	0.106	UI
J_{3u}	162.9.4.7	0.115	UI
Even-odd jitter, pk-pk	162.9.4.7	0.025	UI

^a For a PMD in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c J_{3u} , J_{RMS} , and even-odd jitter measurements are made with a single transmit equalizer setting selected to compensate for the loss of the transmitter package and TP0 to TP0v test fixture.

163.9.2.1 Transmitter test fixture

Unless otherwise noted, measurements of the transmitter are made at the output of a test fixture (TP0v) as shown in Figure 163–3 and described in Annex 163A. An example test fixture is described in Annex 163B.

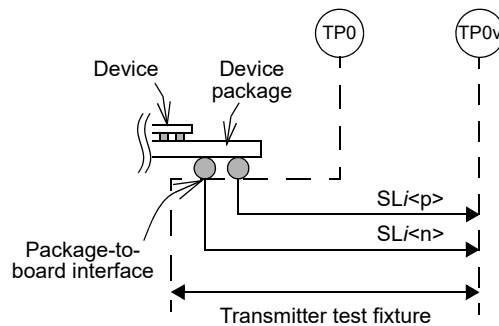


Figure 163–3—Transmitter test fixture and test points

163.9.2.1.1 Test fixture insertion loss

The insertion loss of the test fixture shall be between 1.7 dB and 5 dB at 26.56 GHz. The magnitude of the insertion loss deviation of the test fixture shall be less than or equal to 0.2 dB from 0.05 GHz to 26.56 GHz. Insertion loss deviation is calculated as specified in 93A.4, where T_i is 0.01 ns, and f_b and f_r values are taken from Table 163–11.

163.9.2.1.2 Test fixture effective return loss (ERL)

The ERL of the test fixture at TP0v is computed using the procedure in 93A.5 with the values in Table 163–6. Parameters that do not appear in Table 163–6 take values from Table 163–11.

Table 163–6—Test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

The ERL at TP0v shall be greater than or equal to 15 dB.

163.9.2.1.3 Test fixture common-mode to common-mode return loss

The common-mode to common-mode return loss of the test fixture shall be greater than or equal to 6 dB at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.2 Transmitter difference ERL

The difference ERL of the transmitter at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) in Table 163–5.

Table 163–7—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Tukey window flag	tw	1	—

163.9.2.3 Transmitter common-mode to common-mode return loss

The common-mode to common-mode return loss shall be greater than or equal to RL_{cc} (min) in Table 163–5 at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.4 Difference steady-state voltage

The difference steady-state voltage of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with $N_v = 200$ and other parameter values specified in Table 163–11.

The difference steady-state voltage at TP0v shall meet the specification dV_f (min) in Table 163–5.

163.9.2.5 Difference linear fit pulse peak ratio

The difference linear fit pulse peak ratio of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with parameters specified in Table 163–11.

The difference linear fit pulse peak ratio at TP0v shall meet the specification dR_{peak} (min) in Table 163–5.

163.9.2.6 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is calculated using Equation (163–1).

$$SCMR = 20\log_{10}\left(\frac{V_{peak}}{VCM_{FB}}\right) \quad (163-1)$$

where

$SCMR$	is the signal to AC common-mode noise ratio in dB
V_{peak}	is defined in 162.9.4.1.2
VCM_{FB}	is the full-band peak-to-peak AC common-mode voltage defined by the method specified in 162.9.4.4 and measured with the transmitter equalization set to “no equalization”

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 163–5.

163.9.3 Receiver characteristics

Receiver electrical characteristics are specified at TP5v. The receiver shall meet the specifications given in Table 163–8.

Table 163–8—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	163.9.3.1	53.125 ± 100 ppm	GBd
Difference effective return loss, $dERL$ (min)	163.9.3.3	–3	dB
Differential-mode to common-mode return loss, RL_{cd}	163.9.3.4	Equation (163–2)	dB
Interference tolerance	163.9.3.5	Table 163–9	—
Jitter tolerance	163.9.3.6	Table 162–17	—

163.9.3.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 163.9.3.5 and 163.9.3.6 for any signaling rate in the range specified in Table 163–8.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ± 50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ± 100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).

163.9.3.2 Receiver test fixture

Unless otherwise noted, measurements of the receiver are made at the input of a test fixture (TP5v) as shown in Figure 163–4.

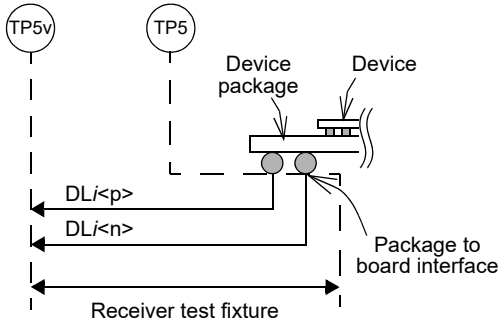


Figure 163–4—Receiver test fixture and test points

Receiver test fixture characteristics are the same as specified in 163.9.2.1.

163.9.3.3 Receiver difference ERL

The difference ERL of the receiver at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP5v to TP5.

The receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 163–8.

163.9.3.4 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (163–2). The differential-mode to common-mode return loss limit is illustrated by Figure 163–5.

$$RL_{cd}(f) \geq \begin{cases} 25 - 20(f/53.125) & 0.05 \leq f \leq 26.5625 \\ 15 & 26.5625 < f \leq 53.125 \end{cases} \quad (163-2)$$

where

$RL_{cd}(f)$ is the differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

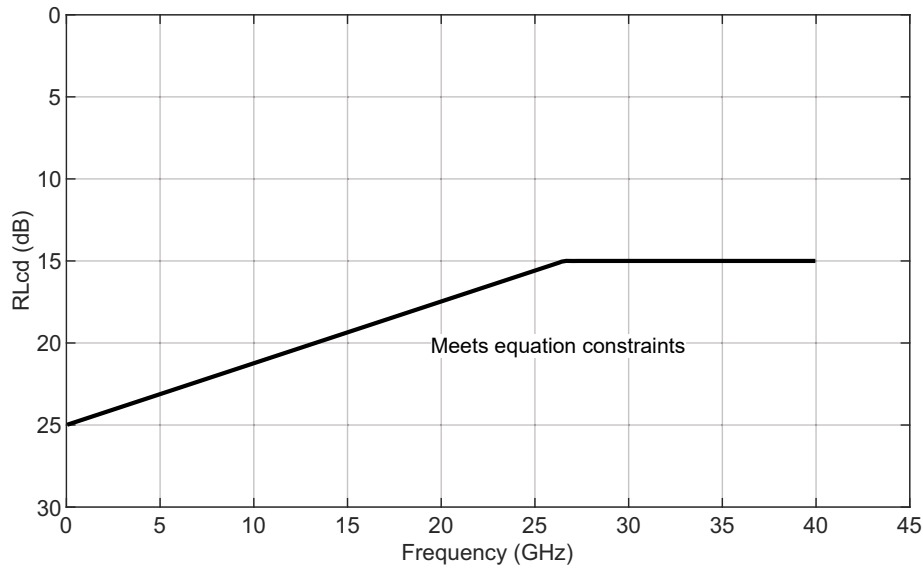


Figure 163-5—Receiver differential-mode to common-mode return loss limit

163.9.3.5 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#). The receiver on each lane shall meet the FEC symbol error ratio requirement with channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 163-9. The following additional considerations apply to the interference tolerance test.

- a) TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A and [Annex 93C](#).
- b) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see [93.8.1.3](#)) is less than or equal to 800 mV.
- c) The ERL of the test setup in [Figure 93C-4](#) measured at TP5 replica towards TPt meets the requirements in [163.10.3](#).
- d) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- e) For the calculation of test channel COM, the transmitter model is determined in one of the following ways.
 - If the transmitter is a device with known S-parameters and transition time T_r , these parameters should be used instead of the transmitter package model in [93A.1.2](#). T_r should be provided as the value at the input of the device S-parameters network, as defined in [120G.3.1.4](#) but with no observation filter.
 - If a calibrated instrument-grade transmitter is used, the TP0 to TP0a trace in [Figure 93C-2](#) and [Figure 93C-3](#) and TP0 to TP0a replica trace in [Figure 93C-4](#) are omitted and the transmitter device package model $S^{(tp)}$ is omitted from [Equation \(93A-3\)](#). The filtered voltage transfer function $H^{(k)}(f)$ calculated in [Equation \(93A-19\)](#) uses the filter $H_f(f)$ defined by [Equation \(93A-46\)](#), where T_r is the measured transmitter transition time (see [120G.3.1.4](#)).
 - If the transmitter is composed of a device with unknown S-parameters or unknown transition time, then the transmitter device package model $S^{(tp)}$ in [93A.1.2](#) is used, and T_r is determined from measurement at TP0v and the TP0 to TP0v S-parameters. The transmitter transition time (see [120G.3.1.4](#)) is measured at TP0v with transmit equalization turned off by setting coefficients to preset 1 values (see [162.9.4.1.3](#)). T_r is set as the value in [Equation \(93A-46\)](#) that

would result in the reference transition time $T_r^{(ref)}$, determined according to 163A.3.1.3 with f_b and A_v equal to values in Table 163–11, being equal to the measured transition time.

- f) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 29$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{3u} and J_{RMS} using Equation (163–3) and Equation (163–4) respectively.
- g) Other COM parameters are set according to the values in Table 163–11.
- h) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 163–11 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- i) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
- j) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
- k) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

$$A_{DD} = \left(\frac{J_{3u}}{2} + Q3d \sqrt{\left((Q3d^2 + 1) \times J_{RMS}^2 - \left(\frac{J_{3u}}{2} \right)^2 \right)} \right) / (Q3d^2 + 1) \quad (163-3)$$

$$\sigma_{RJ} = \left(\frac{J_{3u}}{2} - A_{DD} \right) / (Q3d) \quad (163-4)$$

where

$$Q3d = 3.0902$$

NOTE 1— $Q3d$ is an approximated solution of $Q(Q3d) = 1 \times 10^{-3}$, where the Q function is defined in Equation (95–1).

NOTE 2—Calculation of A_{DD} requires that $(Q3d^2 + 1) \times J_{RMS}^2 \geq \left(\frac{J_{3u}}{2} \right)^2$. If this does not hold, a different transmitter should be used in the test setup.

Table 163–9—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-3}	—	—	10^{-3}	—	—
Insertion loss, IL_{dd} , at 26.5625 GHz ^b	13.5	14.5	—	27.5	28.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

163.9.3.6 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A, Annex 93C, and Annex 120D. The test channel meets the insertion loss requirement for Test 2 in Table 163–9. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 163.9.3.5, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items 3) through 7) in 93C.2, is at least 3 dB.
- For the COM parameter calibration described in 93C.2 item 7), the test channel transmitter J_{RMS} and J_{3u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 163–9 for each case in Table 162–17.

163.10 Channel characteristics

Table 163–10 provides a summary of the channel characteristics for 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, and references to the subclauses addressing each parameter.

163.10.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1 with the values in Table 163–11, where T_r is 7.5 ps for $H_t(f)$ as used in Equation (93A–19). COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2.

Table 163–10—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	163.10.1	3	dB
Maximum insertion loss, IL_{dd} , at 26.56 GHz (recommended)	163.10.2	28	dB
Minimum channel ERL	163.10.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	163.10.4	Equation (163–6)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	163.10.5	Equation (163–7)	dB
Common-mode to differential-mode insertion loss, IL_{dc}	163.10.6	Equation (163–8)	dB
Maximum AC-coupling 3 dB corner frequency	163.10.7	50	kHz

Table 163–11—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	1.2×10^{-4}	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1	z_p	12	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Transmission line characteristic impedance	Z_c	87.5	Ω
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line 2 characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient	$c(-3)$	—	—
Minimum value		–0.06	
Maximum value		0	
Step size		0.02	
Transmitter equalizer, 2 nd pre-cursor coefficient	$c(-2)$	—	—
Minimum value		0	
Maximum value		0.12	
Step size		0.02	

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	–0.34 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	–0.2 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	–20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	–6 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	12	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to 6 $n = 7$ to N_b	$bb_{\max}(n)$	0.85 0.3 0.2 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 –0.03	—
Number of DFE floating tap banks	N_{bg}	3	—
Number of DFE floating taps per bank	N_{bf}	3	—
DFE maximum span including floating taps	N_f	40	UI
Normalized coefficient magnitude limit for DFE floating taps	$b_{g\max}$	0.05	—
DFE floating tap tail root-sum-of-squares limit	$\sigma_{t\max}$	0.02	—

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
DFE floating tap tail starting position	N_{ts}	25	—
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	8.2×10^{-9}	V ² /GHz
Target detector error ratio	DER_0	10^{-4}	—

163.10.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (163–5).

$$ILdd(f) \leq \begin{cases} 0.693 + 2.161\sqrt{f} + 0.607f & 0.01 \leq f \leq 26.5625 \\ -19.12 + 1.773f & 26.5625 < f \leq 40 \end{cases} \quad (163-5)$$

where

$ILdd(f)$ is the insertion loss in dB at frequency f
 f is the frequency in GHz

The insertion loss limit is illustrated by Figure 163–6.

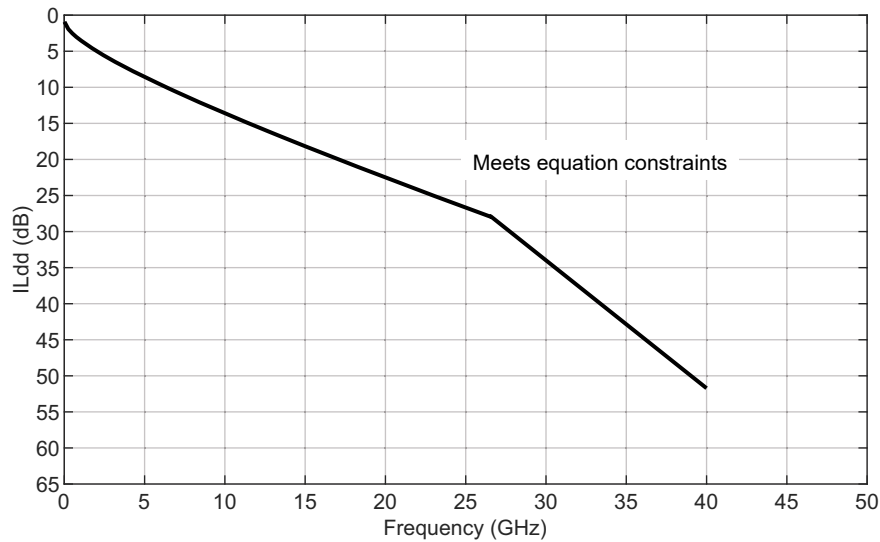


Figure 163–6—Channel insertion loss limit

163.10.3 Channel ERL

The ERL of the channel at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 163–12 and Table 163–11.

Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 163–12—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	3500	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

163.10.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6).

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (163-6)$$

where

$RLcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The TP0 to TP5 channel differential-mode to common-mode return loss limit is illustrated in Figure 163–7.

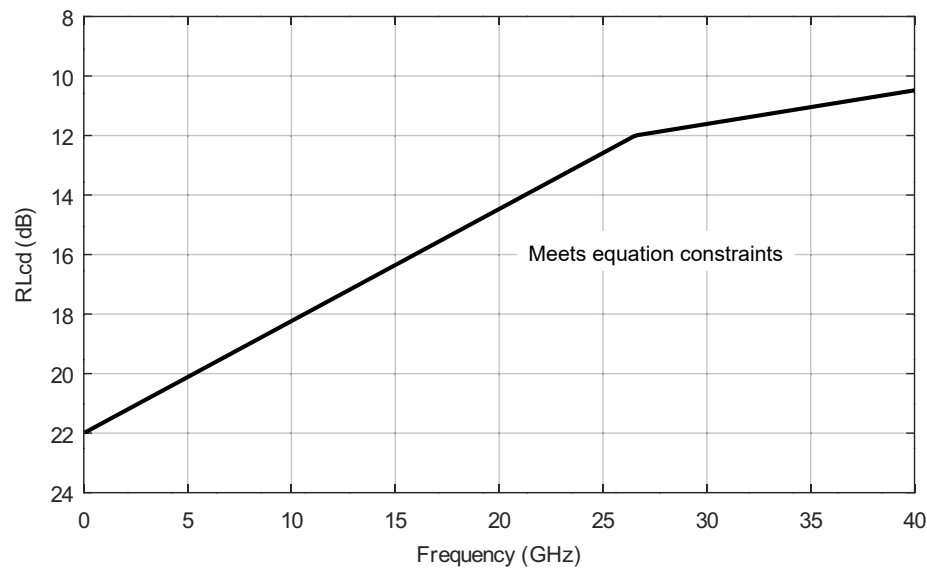


Figure 163-7— Channel differential-mode to common-mode return loss

163.10.5 Channel differential-mode to common-mode insertion loss

The difference between the TP0 to TP5 channel differential-mode to common-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163-7) as illustrated in Figure 163-8.

$$ILcd(f) - ILdd(f) \geq \left\{ \begin{array}{ll} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{array} \right\} \tag{163-7}$$

where

$ILcd(f)$

is the TP0 to TP5 channel differential-mode to common-mode insertion loss at frequency f in dB

$ILdd(f)$

is the TP0 to TP5 insertion loss at frequency f in dB

f

is the frequency in GHz

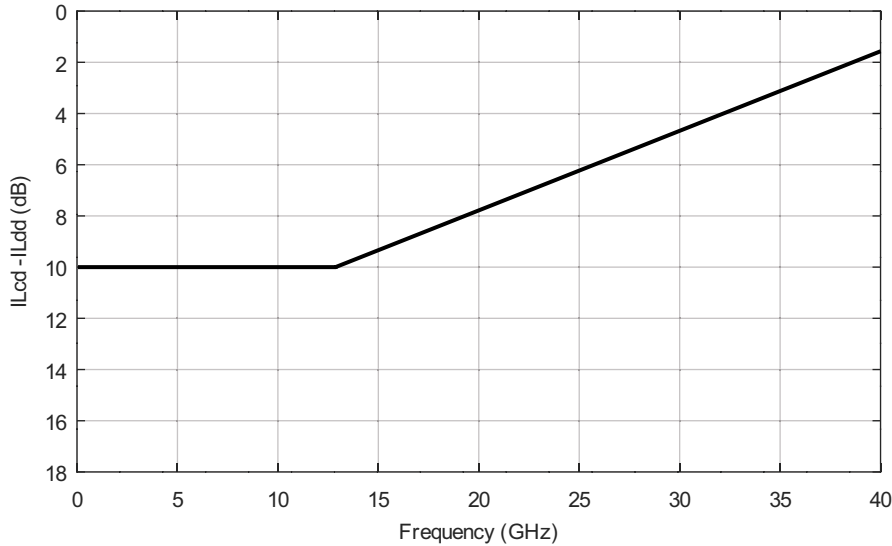


Figure 163-8—Channel differential-mode to common-mode insertion loss to insertion loss difference

163.10.6 Channel common-mode to differential-mode insertion loss

The difference between the TP0 to TP5 channel common-mode to differential-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163-8) illustrated in Figure 163-9.

$$ILdc(f) - ILdd(f) \geq \begin{cases} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{cases} \quad (163-8)$$

where

- $ILdc(f)$ is the TP0 to TP5 channel common-mode to differential-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB
- f is the frequency in GHz

163.10.7 AC-coupling

AC-coupling shall be implemented within the channel (between TP0 and TP5) using DC blocking capacitors. The low-frequency 3 dB cutoff of the channel shall be less than 50 kHz.

Systems with no AC-coupling within the channel are considered engineered links. It is the system integrator's responsibility to verify that the transmitter and the receiver are compatible with the common-mode voltage differences that may exist in this configuration.

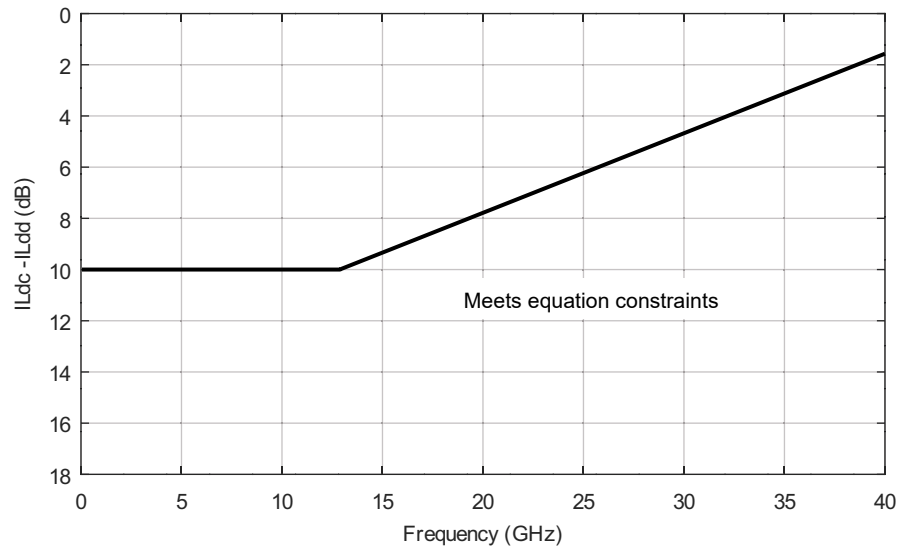


Figure 163-9—Channel common-mode to differential-mode insertion loss to insertion loss difference

163.11 MDI specifications

The MDI for the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 163-2.

163.12 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of [93.10](#).

163.13 Protocol implementation conformance statement (PICS) proforma for Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4¹⁹

163.13.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

163.13.2 Identification

163.13.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

163.13.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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¹⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

163.13.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*KR1	100GBASE-KR1 PMD	163.1	Can operate as a 100GBASE-KR1 PMD	O.1	Yes [] No []
*KR2	200GBASE-KR2 PMD	163.1	Can operate as a 200GBASE-KR2 PMD	O.1	Yes [] No []
*KR4	400GBASE-KR4 PMD	163.1	Can operate as a 400GBASE-KR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-R PMA	163.1	Device implements Clause 135 PMA for 100GBASE-P	KR1:M	Yes [] N/A []
PMA200	200GBASE-P PMA	163.1	Device implements Clause 120 PMA for 200GBASE-R	KR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	163.1	Device implements Clause 120 PMA for 400GBASE-R	KR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	163.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	163.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!KR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	163.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	163.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC 100GBASE-P RS-FEC	163.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	KR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	163.1	Device implements Clause 82 100GBASE-R PCS	KR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	163.1	Device implements Clause 119 200GBASE-R PCS	KR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	163.1	Device implements Clause 119 400GBASE-R PCS	KR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	163.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	163.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	163.6	Conforms to constraints	M	Yes []
*MD	MDIO capability	163.7	Device implements Clause 45 MDIO	O	Yes [] No []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CHNL	Channel	163.10	Item marked with CHNL are channel specifications, these items are the only ones applicable to channels and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	163.9.2		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

163.13.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.13.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

163.13.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	136.8.11.4.1	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC9	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []
PC10	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []

163.13.4.3 Transmitter characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	163.9.2	Per Table 163–5	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD not in same package as PCS	163.9.2	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	TP0 to TP0v test fixture	163.9.2.1	Meets specifications	M	Yes []
TC4	Output voltage specifications	93.8.1.3	Per Table 163–5	M	Yes []
TC5	Difference ERL at TP0v	163.9.2.2	Per Table 163–5	M	Yes []
TC6	Common-mode to common-mode return loss	93.8.1.4	Per Table 163–5	M	Yes []
TC7	Difference steady-state voltage	163.9.2.4	Per Table 163–5	M	Yes []
TC8	Difference linear fit pulse peak ratio	163.9.2.5	Per Table 163–5	M	Yes []
TC9	Level separation mismatch ratio	163.9.2	Per Table 163–5	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 163–5	M	Yes []
TC11	Coefficient range	162.9.4.1.5	Per Table 163–5	M	Yes []
TC12	Signal-to-noise-and-distortion ratio	163.9.2	Per Table 163–5	M	Yes []
TC13	Output jitter	162.9.4.7	Per Table 163–5	M	Yes []
TC14	Transmitter signaling rate	163.9.2	Per Table 163–5	M	Yes []
TC15	Low-frequency peak-to-peak AC common-mode voltage	163.9.2	Per Table 163–5	M	Yes []
TC16	Signal to AC common-mode noise ratio	163.9.2	Per Table 163–5	M	Yes []
TC17	Signal-to-residual-intersymbol-interference ratio	163.9.2	Per Table 163–5	M	Yes []

163.13.4.4 Receiver characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	TP5 to TP5v test fixture	163.9.3.2	Meets specifications	M	Yes []
RC2	Difference ERL at TP5v	163.9.3.3	Greater than or equal to –3 dB	M	Yes []
RC3	Input differential-mode to common-mode return loss	163.9.3.4	Meets constraints of Equation (163–2)	M	Yes []
RC4	Interference tolerance	163.9.3.5	PHY meets FEC symbol error ratio requirement in all tests defined in Table 163–9	M	Yes []
RC5	Jitter tolerance	163.9.3.6	PHY meets FEC symbol error ratio requirement in all cases defined in Table 162–17	M	Yes []

163.13.4.5 Channel characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	ERL	163.10.3	Greater than or equal to 9.7 dB	CHNL:M	Yes [] N/A []
CC2	Channel Operating Margin (COM)	163.10	Greater than or equal to 3 dB	CHNL:M	Yes [] N/A []
CC3	AC-coupling	93.9.4	Between TP0 and TP5, 3 dB cutoff frequency less than 50 kHz	CHNL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CC4	Channel differential-mode to common-mode return loss	163.10.4	Per Equation (163–6)	CHNL:M	Yes [] N/A []
CC5	Channel differential-mode to common-mode insertion loss	163.10.5	Per Equation (163–7)	CHNL:M	Yes [] N/A []
CC6	Channel common-mode to differential-mode insertion loss	163.10.6	Per Equation (163–8)	CHNL:M	Yes [] N/A []

163.13.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	163.12	Conform to applicable requirements of 93.10	M	Yes []